MATERIAL TRANSFER SIMULATION SYSTEM

E4D PLANNER — USER MANUAL & OPERATIONAL REFERENCE GUIDE

Version 2.4 | Discrete Volumetric Simulation & Dynamic Haul Routing

SYSTEM PURPOSE: The E4D Planner Material Transfer Simulator is an engineering and logistics simulation environment designed to model bulk earthmoving, stockpile buffering, and haulage logistics across complex civil and mining projects. It combines an interactive cardinal routing canvas with a daily discrete simulation engine enforcing volume conservation, swell factors, and dynamic overflow throttleback constraints.

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1. System Architecture & Node Topology

The simulation model is constructed from three fundamental node types connected by directed haulage routes:

- **Excavation Nodes (Sources):** Active cut faces, quarry pits, or bulk borrow areas. Materials are excavated in in-situ bulk bank cubic meters (BCM) and transferred downline. Excavations steadily deplete over the 30-day simulation cycle.
- **Stockpile Nodes (Buffers):** Intermediate storage pads and buffer stockpiles (e.g., Stockpile A, Stockpile B). Stockpiles have defined starting volumes and strict maximum capacity ceilings. They handle inbound loose receipts and outbound reclaimed compaction.
- **External Destination Nodes (Sinks):** Final placement locations including offsite processing mills, engineered fills, or commercial sale terminals. These nodes possess effectively unbounded capacity.



Figure 1.1: Stockpile Container Node Card displaying current volume, capacity threshold, and active bulk factors.

2. Graphical User Interface Overview

The workspace is structured into five intuitive operational zones designed for high productivity:

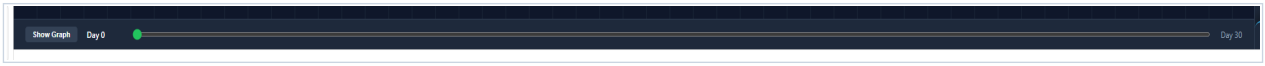


Figure 2.1: Top Global Control Bar featuring Import Model, Export Model, PDF Manual Download, and Playback Toggle.

- **Top Header Bar:** Provides global actions including CSV Model Import, Comprehensive Export dialog, User Manual downloads (PDF and ODT), and the primary "Run Simulation & Play" toggle.
- **Batch Preset Bar:** Positioned directly below the header. Enables one-click application of standard material swell presets (1.15x gravel/clay, 1.25x rock), capacity buffers, and shift multipliers.
- **Central Canvas Viewport:** Infinite interactive spatial layout space. Supports smooth pan/zoom, magnetic container alignment, and graphical route authoring.
- **Right Inspector Panel:** Contextual properties inspector. Displays volumetric metrics and bulk factors for selected containers, or fleet allocations for transfer routes.
- **Bottom Telemetry & Scrubber Tray:** Active during Playback Mode. Hosts a 30-day timeline scrubber with speed controls and a real-time multi-series volume trends graph.

[IMAGE PLACEHOLDER: FULL APPLICATION WORKSPACE OVERVIEW]

Recommended Dimensions: 1920x1080

Figure 2.2: Overall Application Workspace with active project model and configuration panels.

3. Container Management & Canvas Mechanics

Containers represent physical site features and can be manipulated freely on the canvas:

- **Creating & Positioning:** Add containers via the toolbar and reposition them by dragging anywhere on their body. Connected transfer links automatically update geometry in real time.
- **Container Halo Highlight:** When a container is selected, it receives a wide, luminous outer halo with an opacity factor of 0.52. This soft halo is positioned behind the container card, ensuring that the simulation engine's status border (green, amber, red) remains 100% visible and uncompromised.
- **Multi-Selection:** Hold Ctrl (or Cmd on macOS) while clicking containers to select multiple facilities for synchronized batch updates or comparative telemetry tracking.



Figure 3.1: Excavation Source Node Card showing in-situ reserves and initial volume.

4. Intelligent Proximity Snapping (12px Gap & 24px Threshold)

To maintain clean, professional site layouts without requiring tedious manual pixel adjustments, the canvas features an intelligent proximity snapping engine:

- **Magnetic Proximity Range (24px):** When dragging a container, if any of its edges come within 24 pixels of a neighboring container boundary, magnetic snapping activates.
- **Target Separation Margin (12px):** Upon snapping, the container automatically aligns to maintain an exact 12px visual gap between adjacent edges.
- **Axis-Isolated Edge Snapping:** Snapping acts independently on each coordinate axis. Bringing a container vertically downward towards another snaps its top/bottom edges into a 12px gap without displacing horizontal alignment, unless the horizontal edges are also independently within the 24px threshold.

[IMAGE PLACEHOLDER: MAGNETIC SNAPPING ALIGNMENT GUIDE]

Recommended Dimensions: 800x400

Figure 4.1: Side-specific snapping behavior maintaining clean 12px spacing without cross-axis interference.

5. Transfer Link Routing, Corridor Separation & Rate Badges

Material transfer routes define directed haulage corridors between nodes:

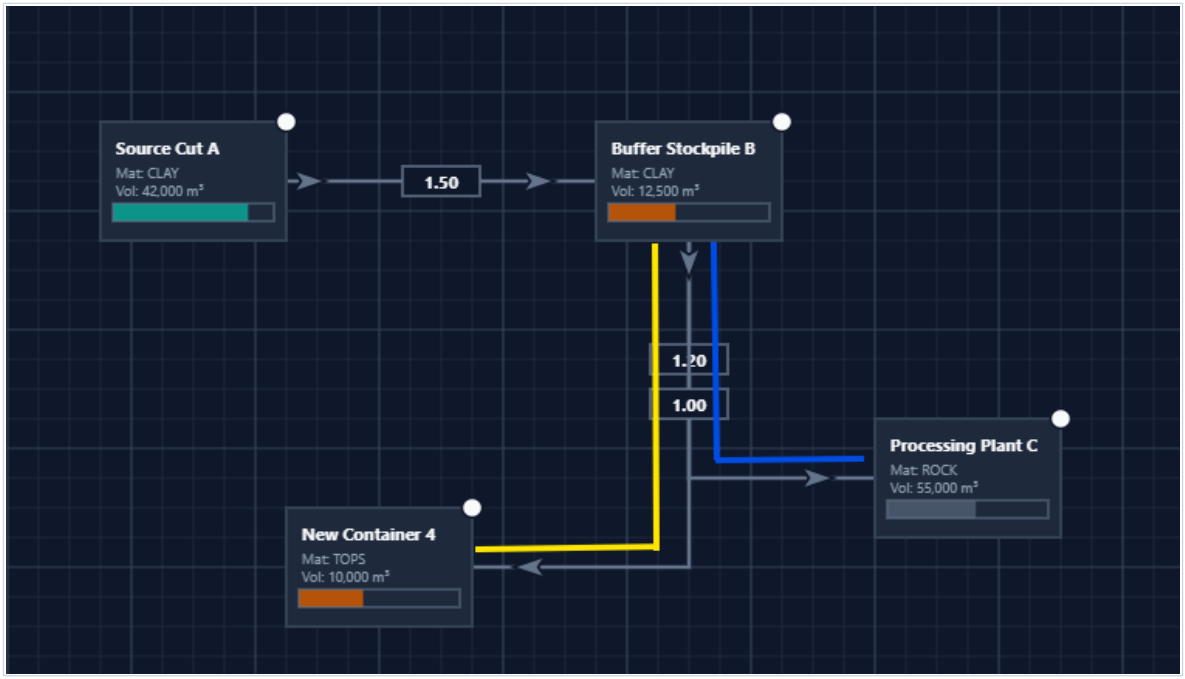


Figure 5.1: Multi-link corridor routing showing parallel lane offsets and flow rate badges.

5.1 Manhattan Orthogonal Routing & 12px Corridor Separation

Routes automatically negotiate 90-degree orthogonal bends between container boundary ports. When multiple links traverse a common spatial corridor (such as concurrent inputs and outputs at a buffer stockpile), the engine automatically separates routes into distinct parallel lanes with a strict 12px corridor separation, preventing any overlapping lines.

5.2 Dynamic Flow Rate Badges

Every active transfer link displays a prominent pill badge displaying its planned or simulated daily flow rate in standardized thousands-notation ('k' notation, e.g. 1.50k for 1,500 m³/day):



Figure 5.2: Flow rate badges displayed along route segments in clean thousands notation.

5.3 Operational Route Status Color Codes

Status Indicator	Operational Definition & Engine Response
GREEN (Full Flow)	100% of planned material volume is moving. Upstream source has ample reserves, haul fleet is fully utilized, and downstream receiver has adequate headroom.
AMBER (Restricted)	Flow is throttled (1% to 99% of planned rate). Triggered when downstream stockpile approaches capacity and engages the Throttleback rule, or when source nears depletion.
RED (Blocked / Halt)	0% flow. Receiving stockpile is 100% full (overflow hold) or upstream source is completely exhausted. Haulage halts completely along this corridor.
BLUE / SLATE (Idle)	Inactive route or unsimulated route segment outside active scheduling window.

6. Volumetric Simulation Engine & Overflow Rules

The simulation engine calculates material movement in discrete 24-hour cycles across a 30-day projection envelope, enforcing physical laws of mass and volume:

6.1 Volumetric Swell Ratios

Material in-situ expands upon excavation into loose cubic meters (LCM) by the bulkInFactor (typically 1.15x to 1.30x). When stockpiled material is reclaimed and compacted into engineered fills, the bulkOutFactor is applied. The simulation guarantees strict volumetric balance across all transfers.

6.2 Overflow Rule: Throttleback

When a downstream stockpile is projected to exceed its maximum capacity on Day t, the simulation engine engages Upstream Throttleback:

- **1. Headroom Evaluation:** AvailableSpace = CapacityMax - CurrentVolume.
- **2. Inflow Capping:** Inbound material volume is capped exactly to match AvailableSpace plus any outbound discharge occurring on the same day.
- **3. Rate Restriction:** Inbound route rates are curtailed proportionally, transitioning link indicators from green to amber.
- **4. Source Preservation:** Upstream excavation nodes retain un-excavated material in bank volume, eliminating phantom material loss.

6.3 Overflow Rule: Headroom Hold

If a buffer stockpile reaches critical capacity with zero outbound relief, the simulation locks the container into a Headroom Hold state. Inbound haulage routes turn red, and backpressure propagates up the supply chain, alerting site engineers to assign additional hauling assets or expand offsite disposal capacity.

7. Simulation Playback & Interactive Telemetry Graph

Clicking "Run Simulation & Play" executes the 30-day projection and transforms the interface into an interactive operational review deck:

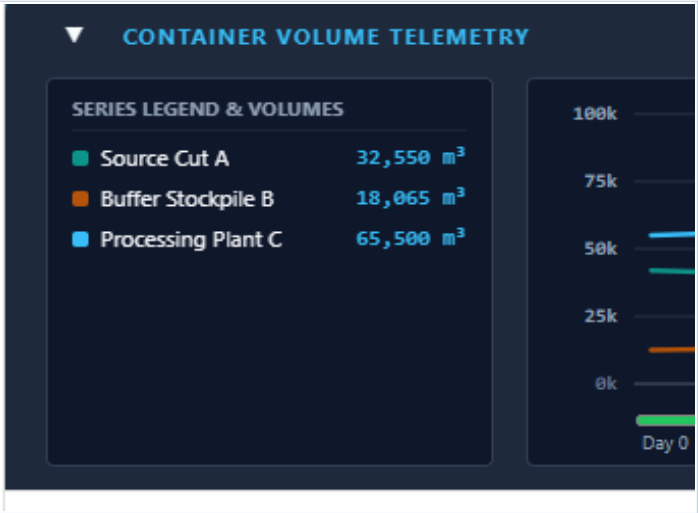


Figure 7.1: Telemetry Graph panel showing multi-container volume trends and interactive legend toggles.

7.1 Timeline Scrubber & Dynamic Animation

The bottom timeline controls allow engineers to step through Day 1 to Day 30 at variable playback speeds (1x, 2x, 5x, 10x). As the timeline advances, container fill gauges update, route lines animate with moving dashed pulses matching flow velocity, and badges show instantaneous daily throughput.

7.2 Interactive Graphline Highlighting & Dual-Halo Visual System

To inspect specific critical paths amidst busy networks, the system implements a coordinated dual-halo highlight system:

- **Legend & Curve Selection:** Clicking any container in the canvas, or clicking its corresponding entry in the telemetry graph legend, activates focused tracking.
- **Graphline Halo (0.42 Opacity):** The selected curve in the graph receives a soft, widened halo beneath the main line, instantly differentiating it from overlapping curves without losing color clarity.

- **Container Card Halo (0.52 Opacity):** Simultaneously, the matching container on the canvas displays a wide luminous halo behind its border, allowing immediate spatial identification across large site layouts.
- **Ctrl + Click Multi-Highlight:** Hold Ctrl or Cmd to highlight two or more containers together, enabling direct visual comparison between interconnected excavations and buffer stockpiles.
- **Visibility Toggles:** Use the eye/checkbox icons in the legend to hide unselected series completely, isolating high-priority bottlenecks.

8. Preset Bar & Batch Configuration

Located below the primary navigation bar, the Batch Preset Bar accelerates scenario planning:

- **Swell Presets:** One-click toggling between Standard Gravel/Clay (1.15x swell), Blasted Rock (1.25x swell), and Saturated Wet Material (1.35x swell).
- **Capacity Buffers:** Quickly scale all stockpile capacities up or down by 10%, 25%, or 50% to test wet weather delays or construction phase shifts.
- **Haul Fleet Multipliers:** Instantly apply shift length adjustments (e.g., 10-hour single shift vs. 24-hour continuous haulage) to all active links.

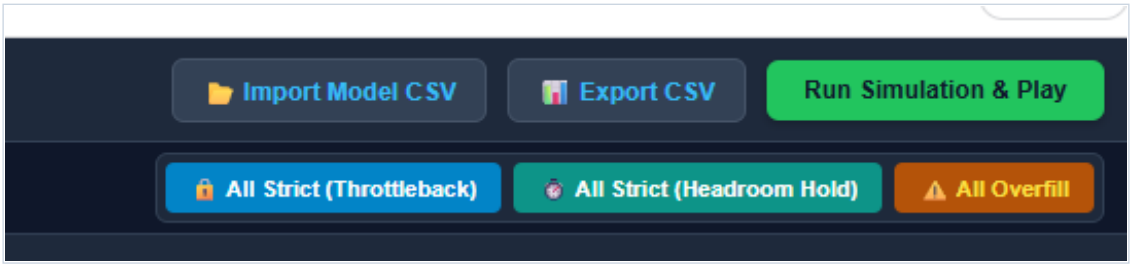


Figure 8.1: Link Inspector and Batch Configuration panel for haul fleet adjustments.

9. CSV Data Model & Export Specifications

All simulation topologies and daily operational outputs are fully accessible through standard CSV import and export.

9.1 Model Structure CSV Specification

CSV Column	Description & Expected Values
record_type	NODE for container definitions, LINK for transfer haulage routes.
id, name, type	Unique string identifier, display label, and entity type (excavation, stockpile, external).
x, y	Canvas pixel coordinates of container top-left corner.
initial_volume, max_capacity	Initial starting volume (m^3) and maximum storage threshold (m^3).
bulk_in, bulk_out	Volumetric expansion factors (e.g. 1.15, 1.00).
source_id, target_id, planned_rate	For LINK records: Origin node ID, destination node ID, and daily nominal haul capacity (m^3/day).

9.2 Simulation Daily Telemetry CSV Specification

Exported simulation audit logs provide complete 30-day discrete accounting:

- **day:** Integer projection day (1 to 30).
- **container_id, container_name:** Target facility identification.
- **start_volume, end_volume:** Beginning and ending day volumes (m^3).
- **total_inflow, total_outflow:** Cumulative daily transfer volumes (m^3).
- **utilization_pct:** Stockpile volume as a percentage of maximum capacity.
- **operational_status:** Operational status code (FULL_FLOW, RESTRICTED, OVERFLOW_HOLD, DEPLETED).

[IMAGE PLACEHOLDER: CSV EXPORT CONFIGURATION MODAL]

Recommended Dimensions: 650x450

Figure 9.1: CSV Export selection modal with Model Schema and Daily Telemetry export buttons.

10. Quick Reference & Operational Shortcuts

Keyboard / Mouse Action	Operation & Result
Left Click Container	Select single container; open node inspector; trigger luminous 0.52 halo.
Ctrl + Click (or Cmd + Click)	Multi-select containers or multi-highlight telemetry graph curves.
Drag Container	Reposition on canvas with automatic 12px gap magnetic edge snapping (within 24px range).
Middle Click / Space + Drag	Pan canvas viewport across infinite layout space.
Mouse Wheel	Zoom canvas in and out centered on cursor.
Click Route Line / Badge	Open route inspector to configure daily haul capacity and truck fleet allocation.
Spacebar (Playback Mode)	Toggle simulation playback between Play and Pause.

> **TECHNICAL SUPPORT & INQUIRIES:** For operational assistance, batch schema customizations, or enterprise system integrations, please contact the E4D Planner Operations Team.